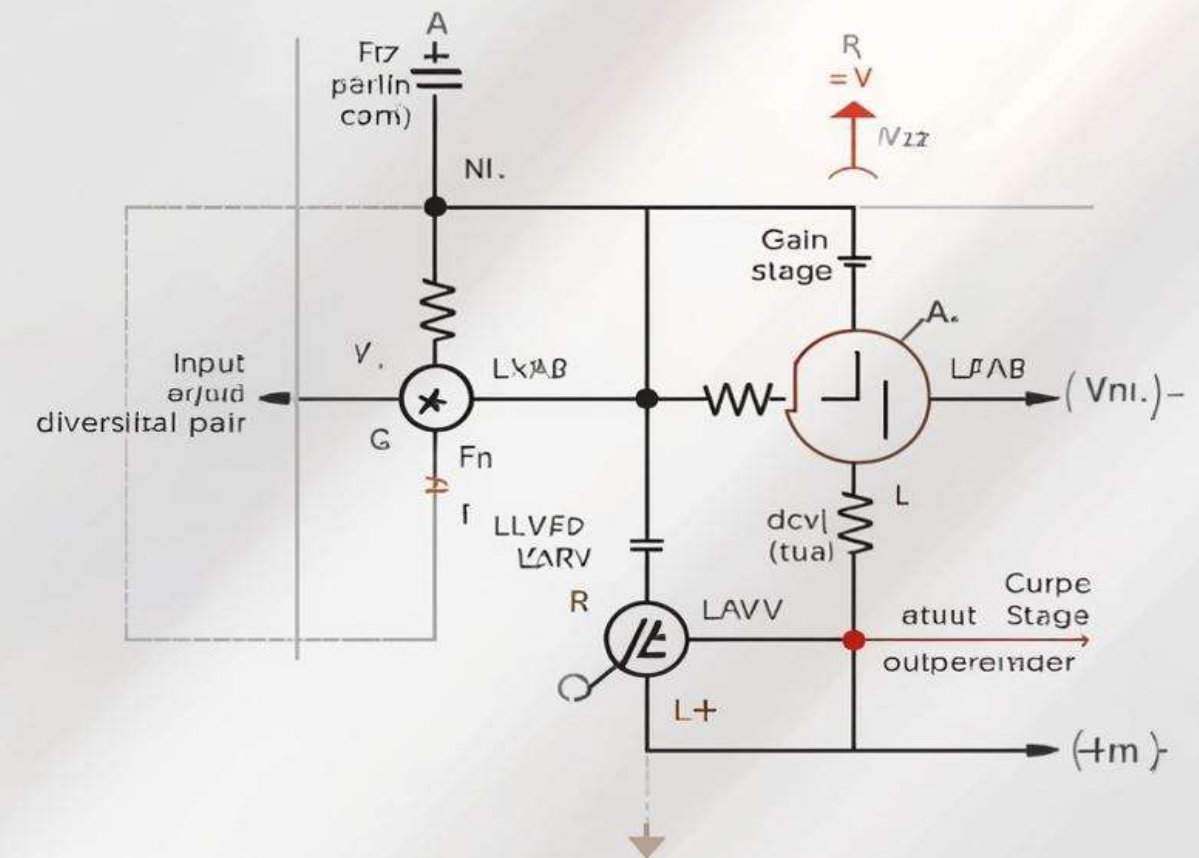
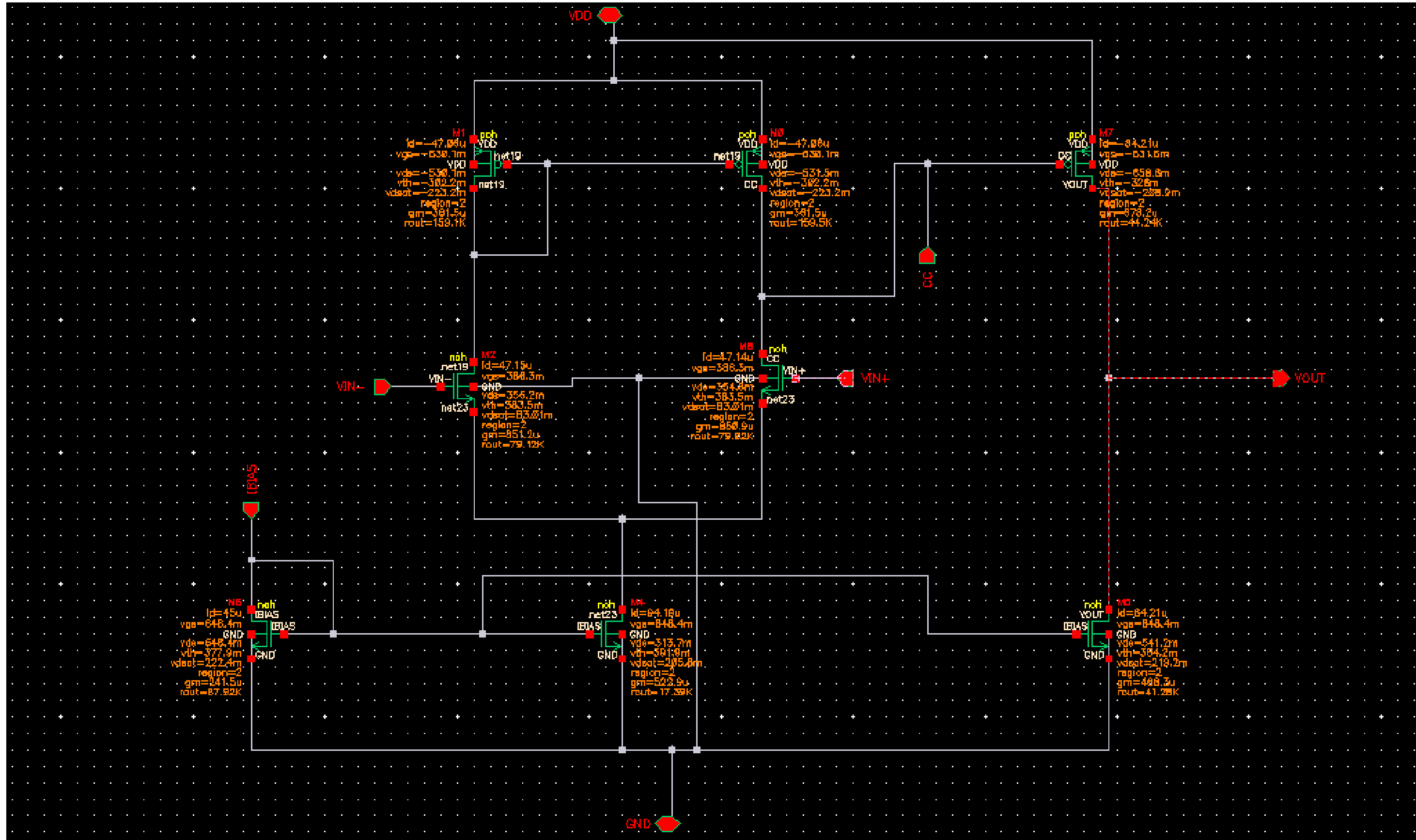


Presented by Yousef Moustafa
Yousef Elsayed

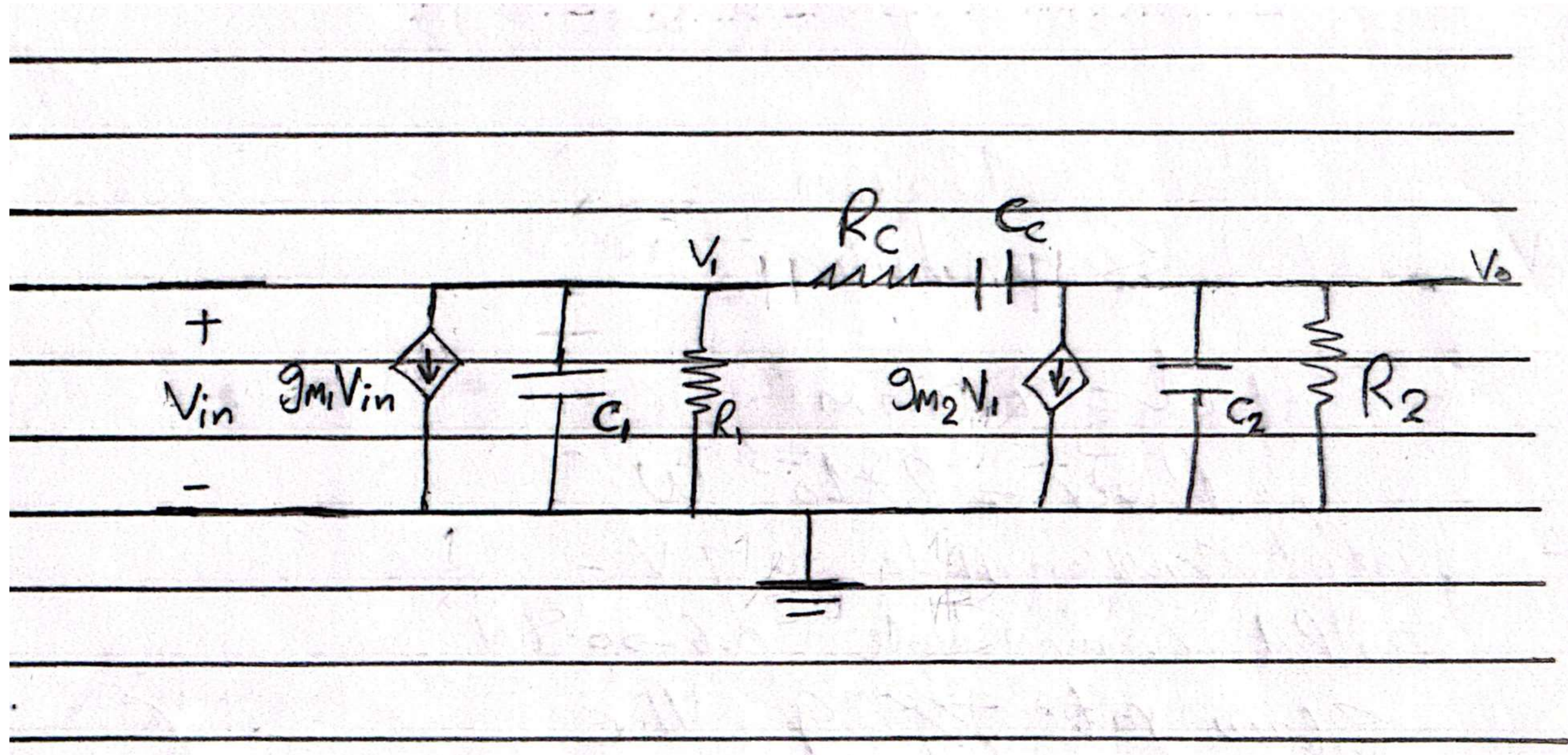


Basic Structure



Hand Analysis

Small signal model



$$\frac{V_i}{1/sC_1} + \frac{V_i}{R_1} + g_{m1} V_{in} + \frac{V_i - V_o}{1/sC_c} = 0$$

$$V_i (sC_1 + 1/R_1 + sC_c) + g_{m1} V_{in} - V_o sC_c = 0$$

$$V_i = \frac{V_o \cdot sC_c R_1 - g_{m1} R_1 V_{in}}{1 + s R_1 (C_1 + C_c)} \Rightarrow \textcircled{1}$$

$$\frac{V_o}{1/sC_2} + \frac{V_o}{R_2} + g_{m2} V_i + \frac{V_o - V_i}{1/sC_c} = 0$$

$$V_o \left[s(C_2 + C_c) + \frac{1}{R_2} \right] = V_i (sC_c - g_{m2})$$

$$V_o \left[s(C_2 + C_c) + \frac{1}{R_2} \right] = V_i (sC_c - g_{m2})$$

$$V_o \left[s(C_2 + C_c) + \frac{1}{R_2} \right] = \frac{(V_o sC_c R_1 - g_{m1} R_1 V_{in})(sC_c - g_{m2})}{1 + s(C_1 + C_2)R_1}$$

$$V_o \left[s(C_2 + C_c) + \frac{1}{R_2} \right] (1 + s(C_1 + C_2)R_1) = \frac{(V_o sC_c R_1 - g_{m1} R_1 V_{in})(sC_c - g_{m2})}{(sC_c - g_{m2})}$$

$$\text{right hand side} \rightarrow V_o (s^2 C_c^2 R_1 - sC_c g_{m2} R_1) - g_{m1} R_1 V_{in} (sC_c - g_{m2})$$

$$V_o \left[(s(C_2 + C_c) + 1/R_2) * (1 + s(C_1 + C_2)R_1) - s^2 C_c^2 R_1 + sC_c g_{m2} R_1 \right] = -g_{m1} R_1 V_{in} (sC_c - g_{m2})$$

$$\text{expand the left side} \rightarrow V_o \left[s(C_2 + C_c) + s^2(C_2 + C_c)(C_1 + C_2)R_1 + \frac{1}{R_2} + s \frac{R_1}{R_2} (C_1 + C_2) \right]$$

$$\underline{V_o} = \frac{g_{m1} R_1 g_{m2} R_2 \left(1 - \frac{s C_c}{g_{m2}}\right)}{1 + s [R_2 (C_1 + C_2) + R_1 (C_c + C_1) + C_c g_{m2} R_1 R_2]}$$

$$+ s^2 R_1 R_2 (C_1 C_2 + C_1 C_c + C_2 C_c)$$

$$\text{* First Pole } P_1 = \frac{1}{R_2 (C_c + C_2) + R_1 (C_c + C_1) + g_{m2} R_2 R_1 C_c}$$

$$P_1 \approx \frac{1}{g_{m2} R_2 R_1 C_c}$$

$$\begin{aligned} \text{* Second Pole } H(s) &= \frac{A_{DC} \left(1 - \frac{s}{z}\right)}{\left(1 + s/P_1\right) \left(1 + s/P_2\right)} \\ &= \frac{A_{DC} \left(1 - s/z\right)}{1 + s \left(1/P_1 + 1/P_2\right) + s^2 \left(1/P_1 P_2\right)} \end{aligned}$$

$$P_1 P_2 = \frac{1}{R_1 R_2 (C_1 C_2 + C_1 C_c + C_2 C_c)}$$

$$P_2 = \frac{g_{m2} C_c}{\underbrace{(C_1 C_2 + C_1 C_c + C_2 C_c)}_{\substack{\text{small} \quad \text{big}}}} \approx \frac{g_{m2}}{C_2}$$

$$\text{* DC gain} = g_{m1} R_1 g_{m2} R_2$$

$$\text{* Zero} = \frac{V_{out1}}{R_c + \frac{1}{s C_c}} \approx g_{m2} V_{out1}$$

$$s_z = \frac{1}{C_c \left(\frac{1}{g_{m6}} - R_2 \right)}$$

* unity Gain Frequency (GBW) $\approx 600 \text{ MHz}$

for a Miller Compensated two stage amp

$$g_m = 2\pi \cdot \text{GBW} \cdot C_c$$
$$g_m = 2\pi \cdot (600 \times 10^6) \times (0.16 \times 10^{-12})$$

$$g_m = 6.032 \times 10^4 \text{ A/V}$$

$$\frac{\omega}{L} = \frac{g_m^2}{2 M_n C_{ox} I_D} = \frac{(6.032 \times 10^4)^2}{300 \times 10^{-6} \times 90 \times 10^{-6}}$$

for 130 nm $\leftarrow$

$$\left(\frac{\omega}{L}\right)_{1.2} \approx 15$$

This is for
 M_1, M_2

as M_1, M_2 determines GBW

$$V_{g1} \leq V_{d1} + V_{t1}$$

$$V_{in} \leq V_{d1} + V_{t1}$$

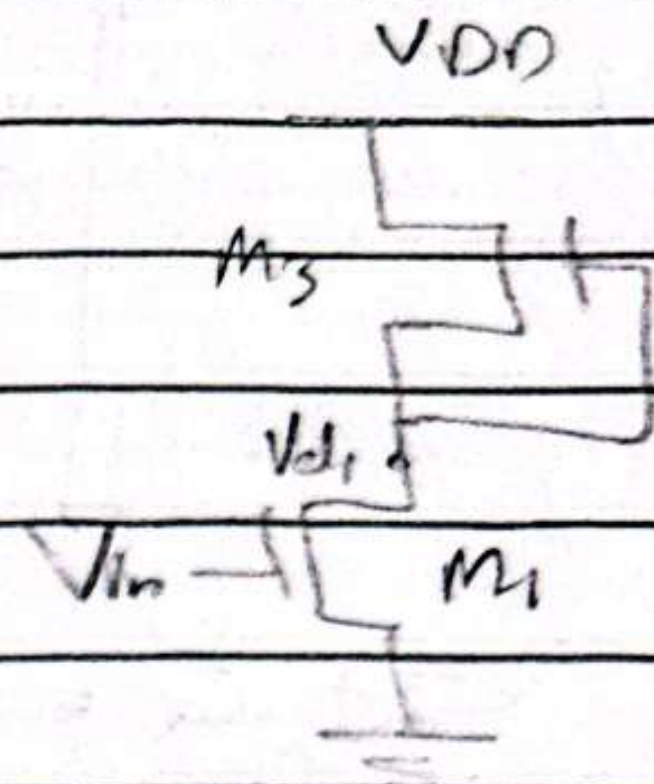
$$V_{d1} = V_{DD} - V_{s3}$$

$$I_3 = \frac{\mu P C_0}{2} \left(\frac{w}{l} \right)_3 (V_{g3} - V_{t3})^2$$

$$V_{s3} = \sqrt{\frac{2 I_3}{\mu P C_0 (w/l)_3}} + V_{t3}$$

$$\rightarrow V_{d1} = V_{DD} - V_{s3}$$

$$V_{in, \max} = V_{DD} - \sqrt{\frac{2 I_3}{\mu P C_0 (w/l)_3}} - V_{t3} + V_{t1}$$



assume $\mu_{pox} = 80 \text{ MA/V}$

$$\text{So } (W/L)_3 = \frac{V_{DD} I_D}{\mu_{pox} (V_{DD} - V_{in} - V_{t3} + V_{t1})^2}$$

$$(W/L)_3 = \frac{90 \times 10^{-6}}{100 \times 10^6 (1.2 - 0.7 - 0.5 + 0.5)^2}$$

$$(W/L)_{3,4} = 4.5 \approx 5$$

$$g_{m4} = 3.67 \times 10^{-4} \text{ A/V}$$

* Current mirror

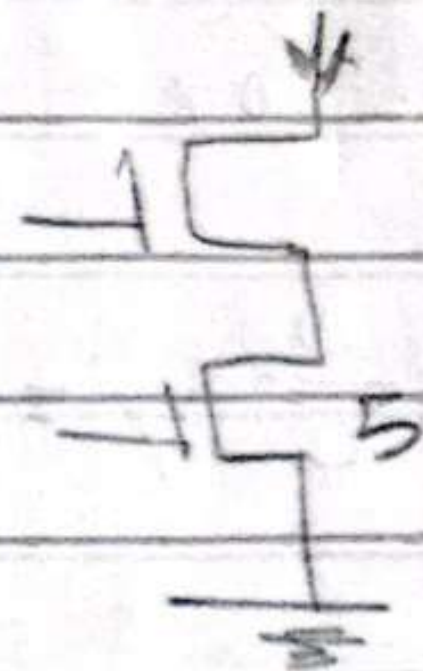
$$V_{in} \geq V_{GS1} + V_{DSat5}$$

$$V_{in} \geq \sqrt{\frac{I_D}{\mu_n C_{ox} (W/L)_3}} + V_t + V_{DSat}$$

$$V_{DSat} \geq V_{in} - \sqrt{\frac{I_D}{\mu_n C_{ox} (W/L)_3}} - V_t +$$

$$V_{DSat} \geq 0.139 \text{ V}$$

$$(W/L)_5 = \frac{2I_{D5}}{\mu_n C_{ox} V_{DSat}} \approx 5$$



* design of Common Source

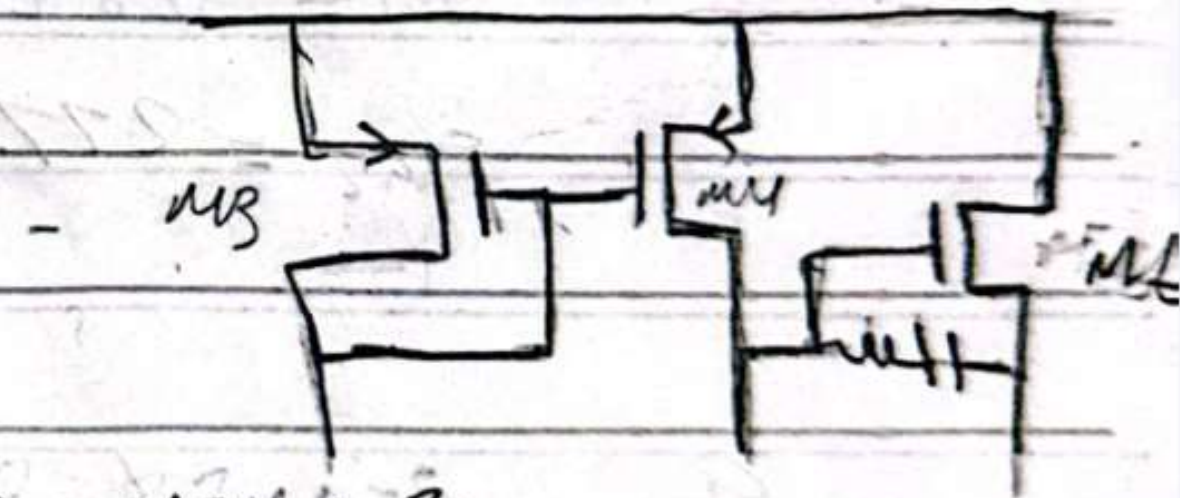
$$GBW = \frac{g_{m1}}{2\pi C_e}$$

for 50° PM $\rightarrow \frac{G_{m6}}{C_L} \approx \frac{4G_{m1}}{C_C}$

$$G_{m1} = 7.5 \times 10^{-3}$$

$$V_{DS3} = V_{DS4} = V_{DS6}$$

$$V_{GS3} = V_{GS4} = V_{GS6}$$

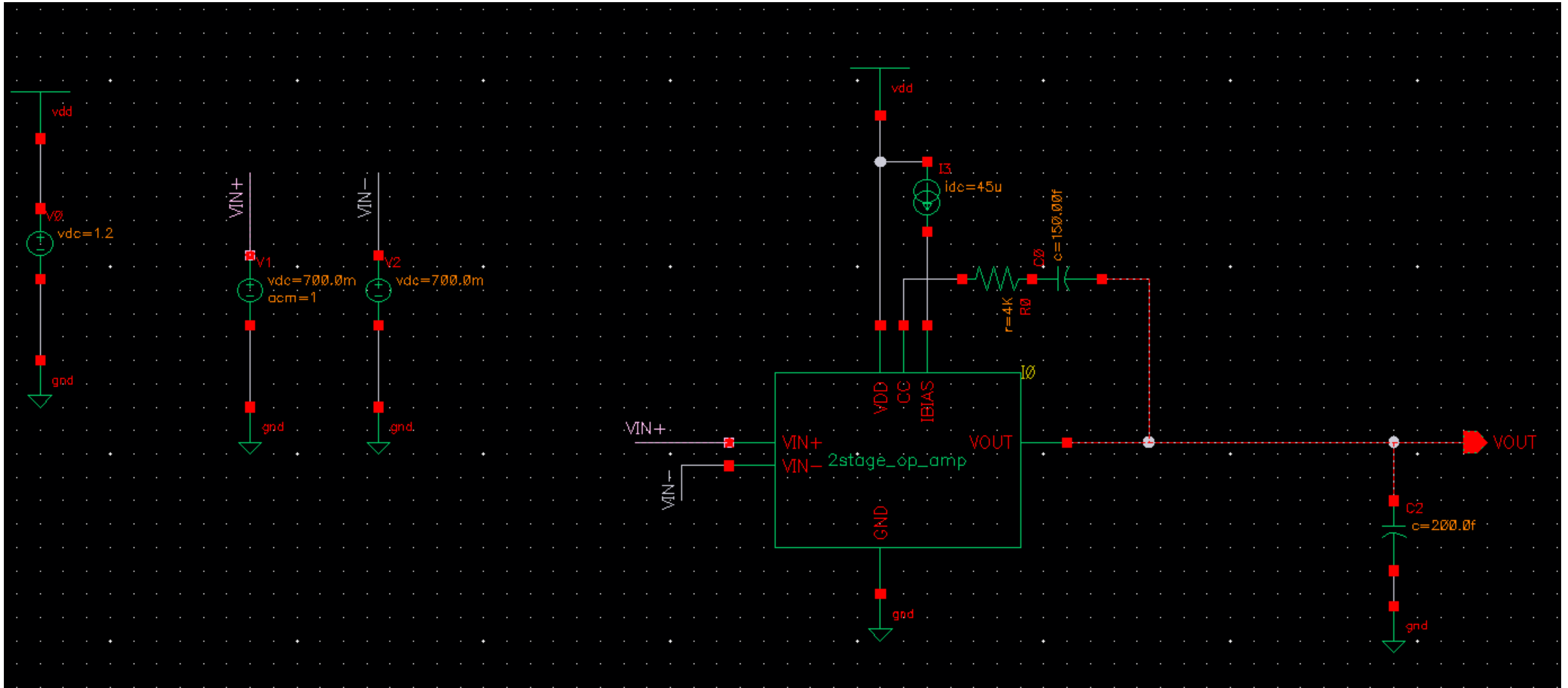


$$G_{m6} \approx \frac{0.2}{0.16} \times 4G_{m1} = 3.0115 \times 10^{-3}$$

$$I_{D4} \approx I_{D6}$$

$$\frac{(w/L)_6}{(w/L)_4} \approx \frac{G_{m6}}{G_{m4}} \rightarrow \left(\frac{w}{L}\right)_6 \approx 9$$

Simulation Result



Power Consumption and Swing

$$* \text{Power dissipation} = 1.2 (94 + 84.1) \times 10^{-6}$$

$$= 2.137 \times 10^{-4} \text{ W} \leq 2 \text{ mW} \checkmark$$

$$* \text{Swing} = 1.2 = 1.2 - (0.1968 - 0.205) = 0.7982 \text{ V}$$

Window Expressions Info Help

cādence

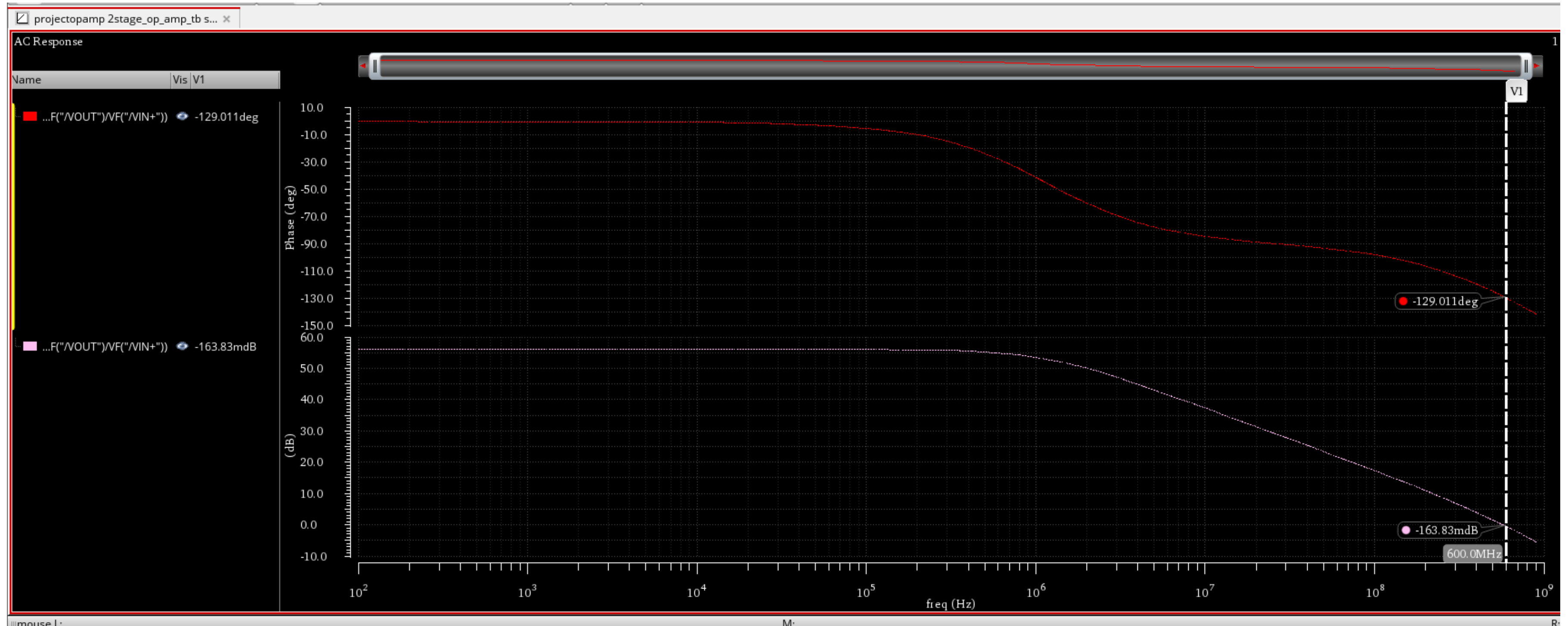
signal	OP("/V0" "??")
--------	----------------

i	-223.4u
---	---------

pwr	-268u
-----	-------

v	1.2
---	-----

Phase margin and DC gain



Phase margin and DC gain

$$Gain = g_{m1} g_{m2} R_{o1} R_{o2}$$

$$g_{m1} = 8500 \mu A/V \quad R_{o1} = 79.82 K \quad R_{o2} = 159.5 K$$

$$g_{m2} = 669.3 \mu A/V \quad R_{o1} = 41.96 K \quad R_{o2} = 45.7 K$$

$$Gain_1 = 8500 \times 10^{-6} \times \left(\frac{79.82 \times 159.5 K}{79.82 + 159.5} \right) \times 10^3 = 45.266$$

$$Gain_2 = 669.3 \times 10^{-6} \times \left(\frac{41.96 \times 45.7}{41.96 + 45.7} \right) \times 10^3 = 14.64$$

$$Gain(dB) = 56.43$$

$$\text{Phase Margin} = 180 - 129 = 51^\circ$$

Untiy gain frequency

Outputs				
	Name/Signal/Expr	Value	Plot	Save
1	ymax(abs(VF("/VOOUT")))	651.6	☒	☐
2	dB20(ymax(abs(VF("/VOOUT"))))	56.28	☒	☐
3	unityGainFreq(VF("/VOOUT"))	600.4M	☒	☐

Output DC-level



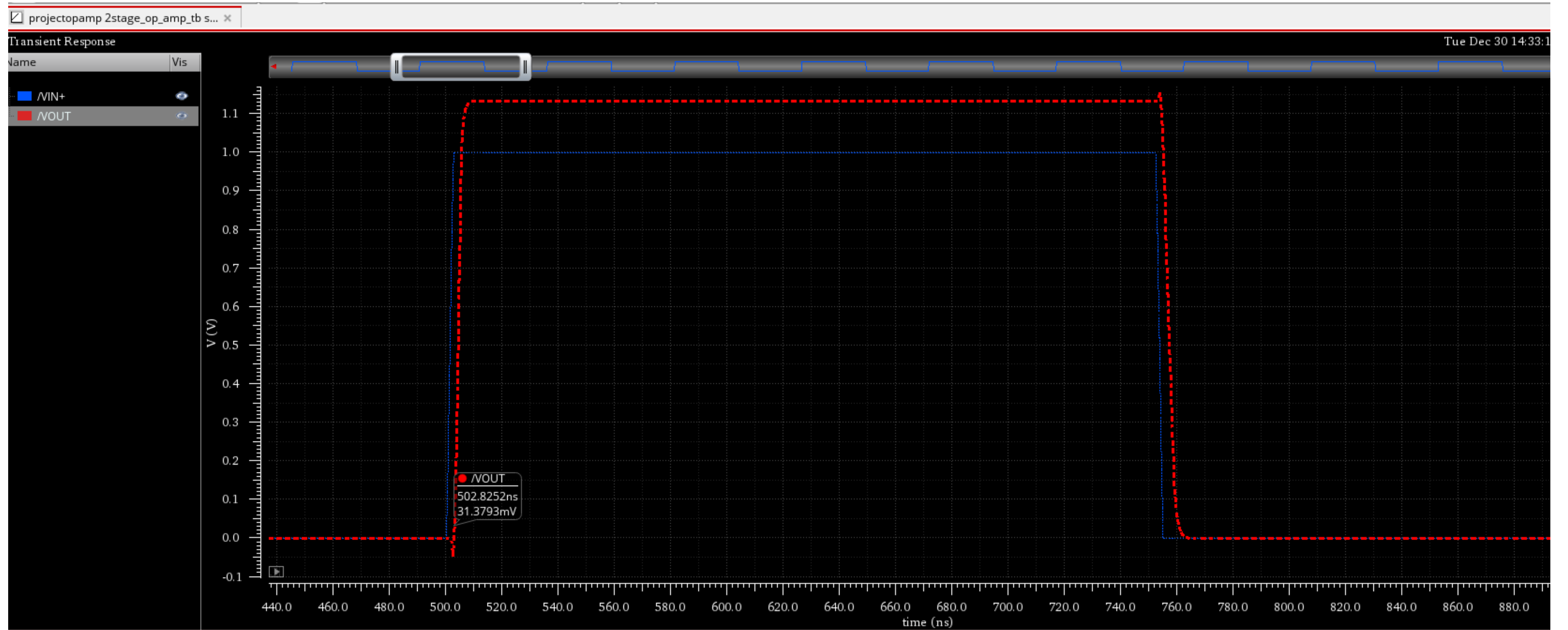
Slew Rate

The slew rate was extracted from the transient response.

We took two different points from the curve and calculated $SR = \Delta V / \Delta t$.

$$\begin{aligned} SR &= \frac{V_2 - V_1}{t_2 - t_1} \\ &= \frac{1.11 - 0.031379}{(506.337 - 502.825) \times 10^{-9}} \\ &= 3.07 \times 10^2 \text{ V}/\mu\text{s} \end{aligned}$$

Slew Rate



**Thank
You**

